AI ACADEMY · PYTHON READINESS BRIDGE · STUDENT LESSON

Values variables and types

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

Before you start

What does print(total) do after total = 24?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Distinguish integer, floating-point, text and Boolean values. Try the recall before opening notes.

Your learning target

I can distinguish integer, floating-point, text and Boolean **value**s.

I can trace **reassignment** without treating a name as a live formula.

I can choose a suitable representation for a count and an identifier.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Distinguish integer, floating-point, text and Boolean **value**s

Trace **reassignment** without treating a name as a live formula

Choose a suitable representation for a count and an identifier

Retrieve what you already know

> What does print(total) do after total = 24?

My starting explanation

This week's end goal

Trace **value**s **variable**s and **type**s and explain the changed program before running it.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Values variables and types case. Separate observations, given facts and assumptions before choosing an explanation.

A count and a shelf label can look similar while needing different operations. The integer 8 is a number; "08" is text that preserves a leading zero.

You can count eight objects, but adding two shelf labels is usually meaningless. This week you will examine **value**s, trace changes to names and explain which representation a task needs.

Use the saved Python 3 script workflow from Week 1. Make a new file **value**s.py, predict before running and keep actual **output** separately. Use fictional labels only.

Four useful **type**s are int for whole numbers, float for floating-point numbers, str for text, and bool for True or False. A **type** describes a **value**, not whether its meaning is sensible for the task.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Values variables and types. Connect each labelled input, deciding step and result.

Literal in code	Type	Possible meaning
8	int	Number of counters
2.5	float	Length in metres
"08"	str	Shelf identifier
True	bool	A yes-or-no condition

```
count = 8
shelf = "08"
```

```
print(type(count))
print(type(shelf))
```

type(count) produces the **type** of the **value** bound to count. These print lines display <class 'int'> and <class 'str'>. The quotes in "08" mark text; they are not displayed by print(shelf).

True and False have capital initials and no quotes. "False" is text, not a Boolean.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Values variables and types. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Trace the five-line count and saved program. Record both printed **output**s, then predict the result of assigning saved = count after the change. Check with a run or label a paper trace.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

> **Change one thing**

Change: Change count to 10, leaving tag unchanged.

Predicted result

Prediction: The two valid **output**s become 12 and 82.

Why the result follows

Reason: Only the integer **variable** changes; the **string** still contains the character 8.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Values variables and types to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Value	A piece of data such as 8 or a text label
Variable	A name bound to a value during execution
Type	A category of value that determines available operations
String	A sequence of text characters
Reassignment	Binding an existing name to a new value

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Values variables and types, predict a step, then compare with its explanation.

A saved value is not a live link

INSTRUCTION	COUNT NOW	SAVED NOW
count = 8	8	Not assigned
saved = count	8	8
count = count + 2	10	8

Evaluate the right side using the current value, then bind the left-hand name. The saved integer does not update when count is reassigned.

```
count = 8
saved = count
count = count + 2
```

```
print(count)
print(saved)
```

The **output**s are 10 and 8 on separate lines. saved = count uses the **value** available at that moment.

It does not store a formula that keeps following count. count = count + 2 is valid assignment: read the old 8, calculate 10, then rebind count. It is not a claim that 8 equals 10.

> **After count becomes 10, add saved = count and print(saved). Predict the new output and explain why this is different from expecting saved to change automatically.**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Values variables and types

A number and its printed-looking text. A program stores a numeric badge count and a **string** that looks like that count.

The facts we will use

> **Supplied facts**
>
> Python 3: count=8; tag="8"; print(count+2); print(tag+"2").

AI ACADEMY / PYTHON-BRIDGE/2

Values variables and types

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Inspect types

count is an integer; tag is a string.





2. Apply addition

Integer addition gives 10; string concatenation gives the text 82.





3. Predict a mixed operation

count+tag raises TypeError rather than silently converting tag.

CHANGE ONE THING

Change count to 10, leaving tag unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Inspect types	count is an integer; tag is a string.
2. Apply addition	Integer addition gives 10; string concatenation gives the text 82.
3. Predict a mixed operation	count+tag raises TypeError rather than silently converting tag.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Values carry **type**s, not merely appearances. The plus operator performs numeric addition for these integers and concatenation for these **string**s; the mixed operation is invalid here.

What this example does not establish

Keep the stated limit

These examples use built-in integers and **string**s; operator behaviour depends on **type**s and is not proof of general model reasoning.

Keep the stated limit

Read the labelled picture: Predict both print **output**s and explain why they differ. Does count+tag work, and what happens to each valid **output** after count becomes ten?

Keep the stated limit

Change one thing: Change count to 10, leaving tag unchanged. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Values variables and types claim and explain a repair using the supplied evidence.

```
print(8 + 2)
print("8" + "2")
print("8" * 2)
```

Check the claim against the facts

These lines display 10, 82 and 88. Number addition calculates a sum. **String** addition joins text; **string** multiplication by a whole number repeats it. Python does not infer from the digits that you wanted a number.

Trying "8" + 2 instead raises **Type**Error because this addition mixes text and an integer.

Check the claim against the facts

For a supplied count, use a numeric literal such as 8. For a shelf ID, preserve text such as "08". Later weeks will parse and validate data read from files; changing a **type** alone will not prove that a **value** is valid.

Use descriptive names with underscores, such as shelf_id. Names are case-sensitive: count and Count are different names.

Locate the earliest unsupported step

Predict 6 + 3, "6" + "3" and "6" * 3. Explain what happens for "6" + 3. Which expression counts six groups of three?

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Values variables and types with support, then explain the deciding step in your own words.

Classify 12, 1.5, "12", False and "False". Explain why the last two are different.

After count becomes 10, add saved = count and print(saved). Predict the new output and explain why this is different from expecting saved to change automatically.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace a saved value

Use workbook W2 for the connected example "**Value**s **variable**s and **type**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Value**s **variable**s and **type**s. Proposed explanation: "Digits always mean a number". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Predict both print **output**s and explain why they differ. Does count+tag work, and what happens to each valid **output** after count becomes ten?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Values variables and types. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Trace the five-line count and saved program. Record both printed **output**s, then predict the result of assigning saved = count after the change. Check with a run or label a paper trace.

```
count = 8
shelf = "08"
print(type(count))
print(type(shelf))
```

```
count = 8
saved = count
count = count + 2
print(count)
print(saved)
```

```
print(8 + 2)
print("8" + "2")
print("8" * 2)
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Values variables and types. Explain why each step is justified.

Explain → apply → challenge

Classify 12, 1.5, "12", False and "False". Explain why the last two are different.

> Predict 6 + 3, "6" + "3" and "6" * 3. Explain what happens for "6" + 3. Which expression counts six groups of three?

Give a suitable **type** for a counter count, a length of 2.5 metres, a shelf ID with a leading zero, and a yes-or-no availability flag. Give a count **value** with the right **type** that should still be rejected.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Python readiness portfolio. Keep an annotated decision and the evidence supporting it.

Add a labelled **value**s **variable**s and **type**s **evidence** sheet to Python readiness portfolio. Show how you can distinguish integer, floating-point, text and boolean **value**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For Python readiness before the Responsible Plant Classifier, retain a trace of the **value**s, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

A negative int is still an integer, but it may be invalid as a count of objects. A correctly **type**d label can still name the wrong shelf. Keep the task's meaning separate from the **value**'s Python **type**.

Week 3 uses arithmetic expressions and grouping to calculate with known numeric **value**s.

A shelf identifier "08" is changed to the number 8 so that it can be added to another identifier. Explain what information is lost and why numeric addition does not solve an identifier task. Give one example of a value with the right type but the wrong meaning.

Give a suitable **type** for a counter count, a length of 2.5 metres, a shelf ID with a leading zero, and a yes-or-no availability flag. Give a count **value** with the right **type** that should still be rejected.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Values variables and types case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

A saved value is not a live link

INSTRUCTION	COUNT NOW	SAVED NOW
count = 8	8	Not assigned
saved = count	8	8
count = count + 2	10	8

Evaluate the right side using the current value, then bind the left-hand name. The saved integer does not update when count is reassigned.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Trace values, types and reassignment

Predict both **output** lines and explain why estimate stays unchanged when count is reassigned. Repeat the trace with initial count=0. Explain why note is text even though its printed characters look like a number.

```
count = 4
note = "4"
estimate = count / 2
fits = count <= 5
count = count + 3
print(count, note, estimate, fits)
print(type(count).__name__, type(note).__name__, type(estimate).__name__,
type(fits).__name__)
```

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A fresh script runs seats = 12; booked = seats; seats = seats - 5; print(seats); print(booked), with each statement on a separate line. Predict both **output**s. Then add booked = seats + 1 and print(booked).

Explain the evaluation order and choose a **type** for a seat label "07".

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Values variables and types, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A number and its printed-looking text. A program stores a numeric badge count and a **string** that looks like that count.

By the end, I can	How I will show it
Distinguish integer, floating-point, text and Boolean values	Explain the deciding step, show a trace or test, and state one limit.
Trace reassignment without treating a name as a live formula	Explain the deciding step, show a trace or test, and state one limit.
Choose a suitable representation for a count and an identifier	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Values variables and types without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Distinguish integer, floating-point, text and Boolean **value**s

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

> block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A number and its printed-looking text. A program stores a numeric badge count and a **string** that looks like that count.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? count+tag raises **Type**Error rather than silently converting tag.

1. Predict a mixed operation

2. Inspect **type**s

3. Apply addition

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Predict a mixed operation count+tag raises **Type**Error rather than silently converting tag. **Value**s carry **type**s, not merely appearances.

The plus operator performs numeric addition for these integers and concatenation for these **string**s; the mixed operation is invalid here.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change count to 10, leaving tag unchanged. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The two valid **output**s become 12 and 82.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The two valid **output**s become 12 and 82. Only the integer **variable** changes; the **string** still contains the character 8.

Repair a missing building block

Review From a paper algorithm to Python (python-bridge/1). Goal: Translate a short paper calculation into a Python script. Short practice: Match each paper quantity to its Python name.

Why does the spare count get added once, and what prints when spare is five?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

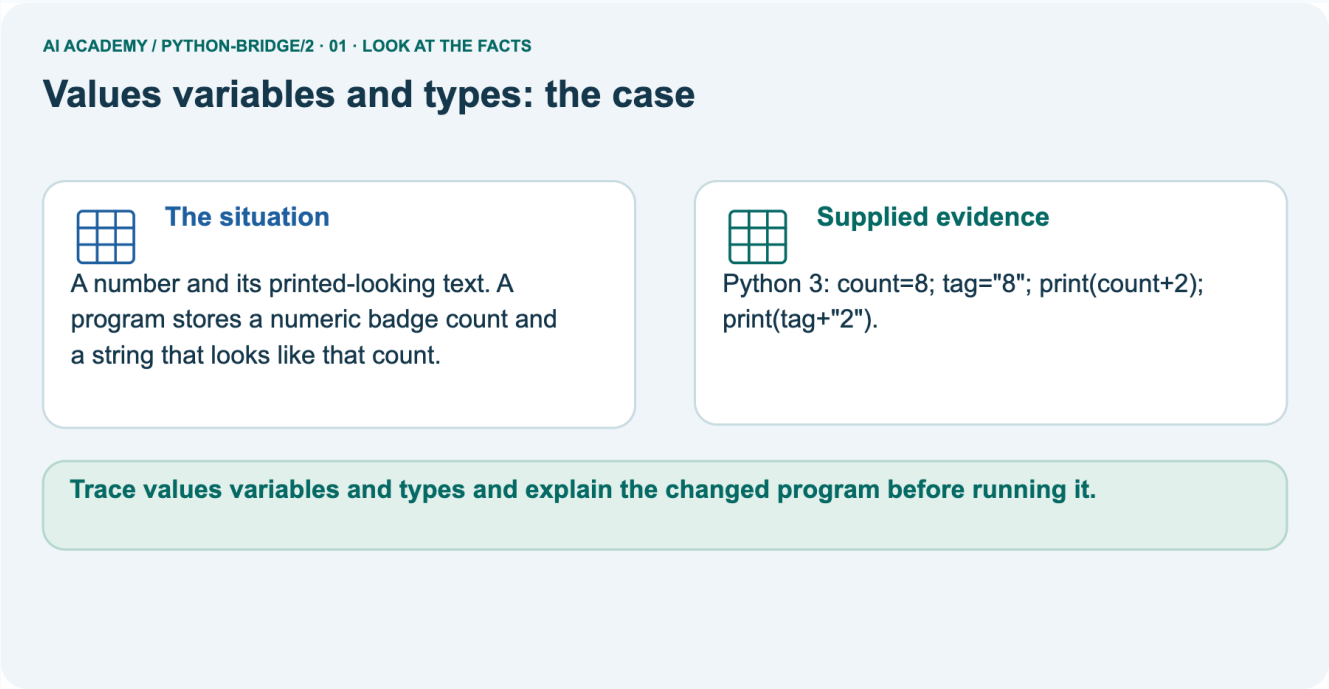
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Values variables and types graphic using its labels, deciding evidence and relevant limitation.

01. Values variables and types: the case

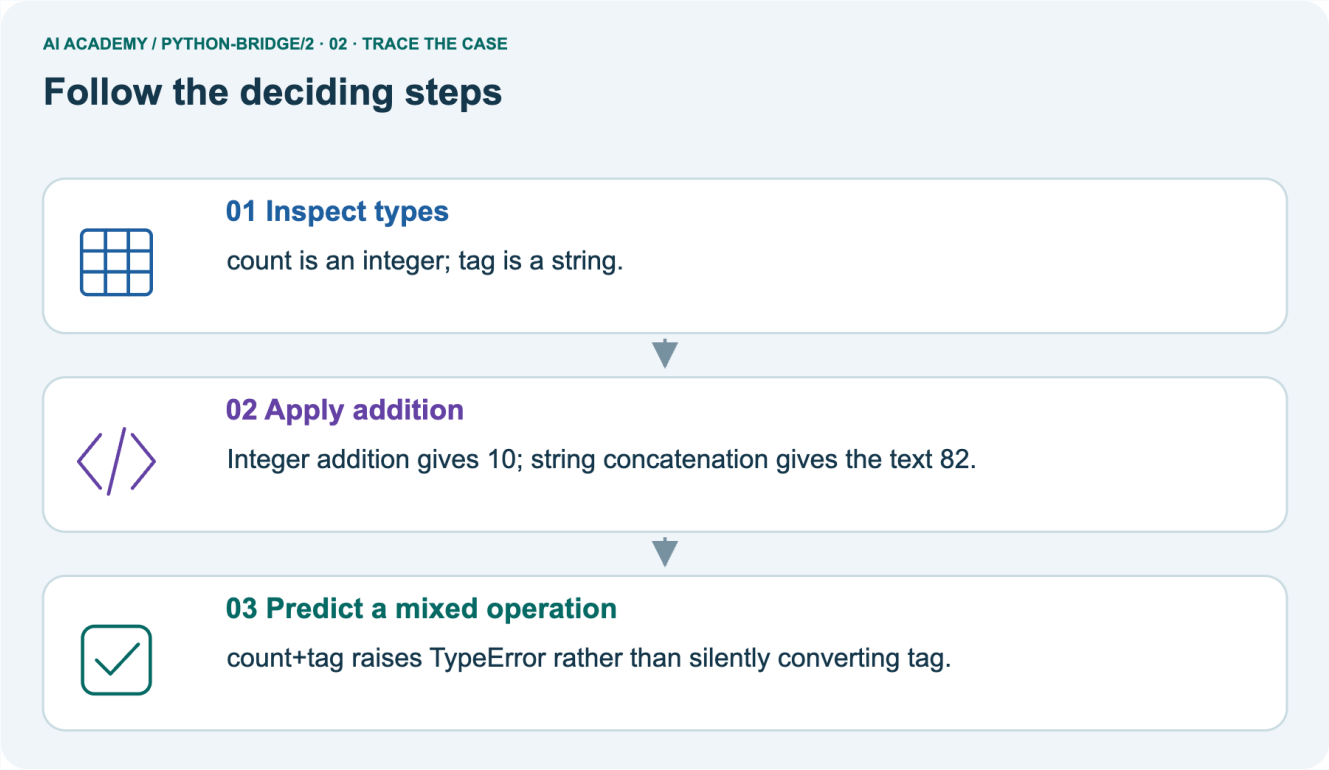


Follow the labelled picture with the matching explanation. Open the vector diagram

A number and its printed-looking text. A program stores a numeric badge count and a **string** that looks like that count. Supplied facts: Python 3: count=8; tag="8"; print(count+2); print(tag+"2").

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

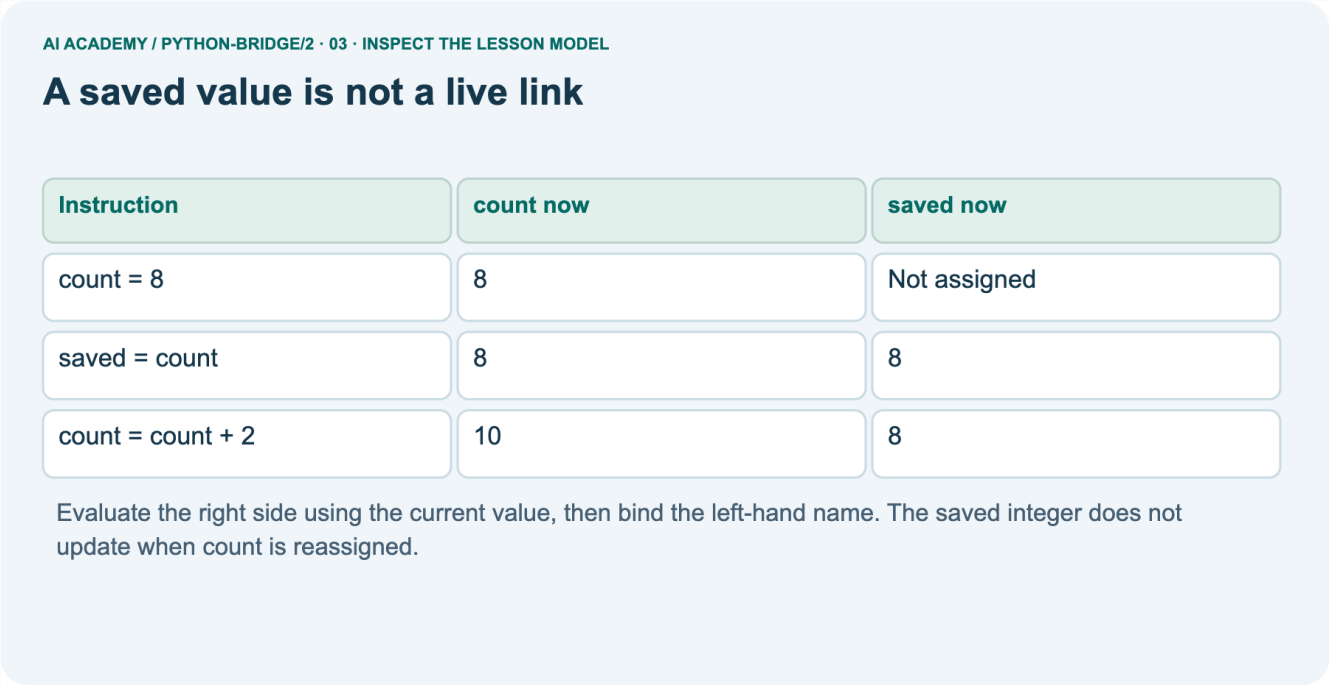


Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect **type**s: count is an integer; tag is a **string**. → Apply addition: Integer addition gives 10; **string** concatenation gives the text 82. → Predict a mixed operation: count+tag raises **Type**Error rather than silently converting tag.

> Point to each numbered stage and explain which recorded **input** it uses.

03. A saved value is not a live link

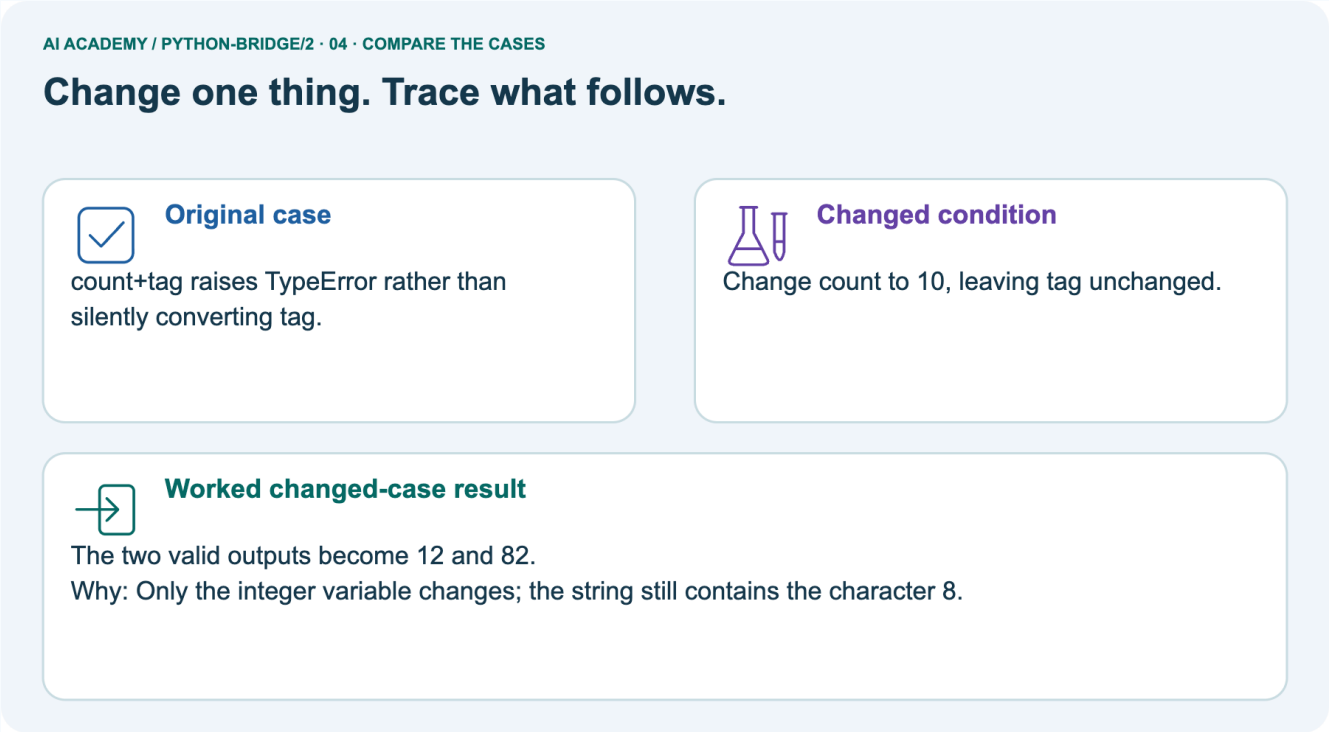


Follow the labelled picture with the matching explanation. Open the vector diagram

A saved **value** is not a live link. count = 8 | 8 | Not assigned. saved = count | 8 | 8. count = count + 2 | 10 | 8

> Read the labels and **value**s. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



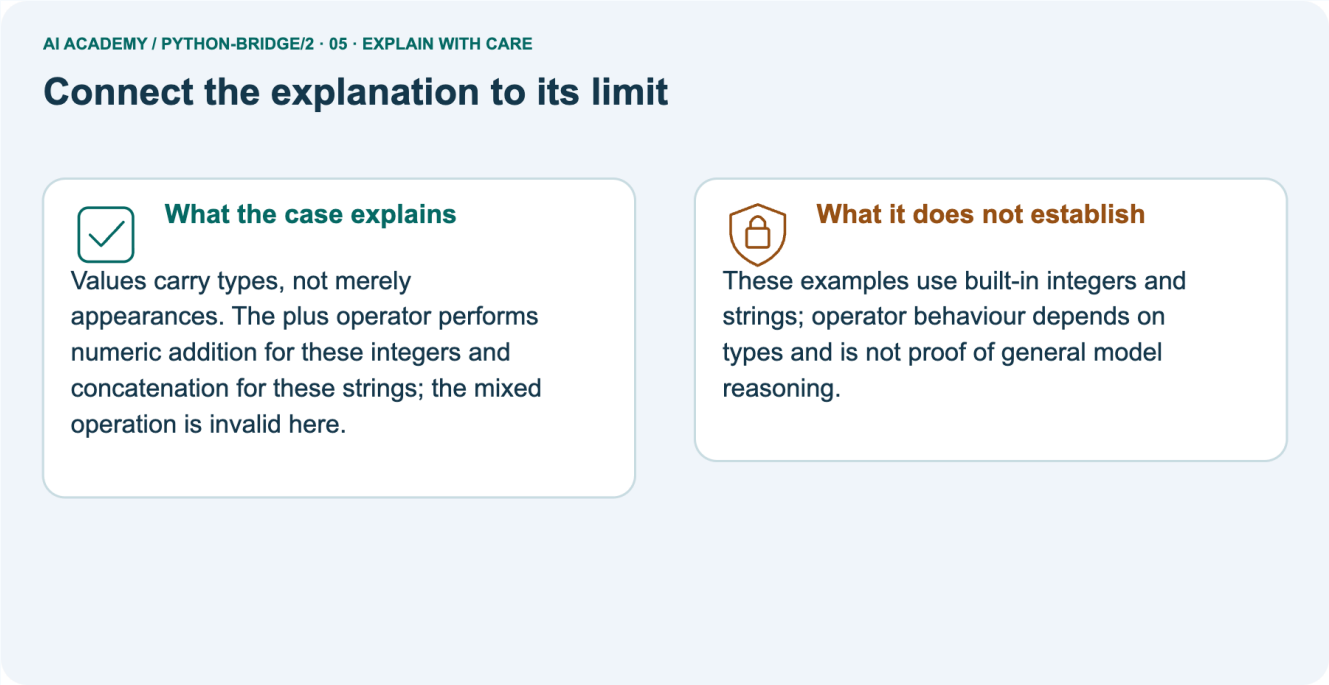
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: count+tag raises **Type**Error rather than silently converting tag. Changed condition: Change count to 10, leaving tag unchanged. Expected result: The two valid **output**s become 12 and 82.

Reason: Only the integer **variable** changes; the **string** still contains the character 8.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



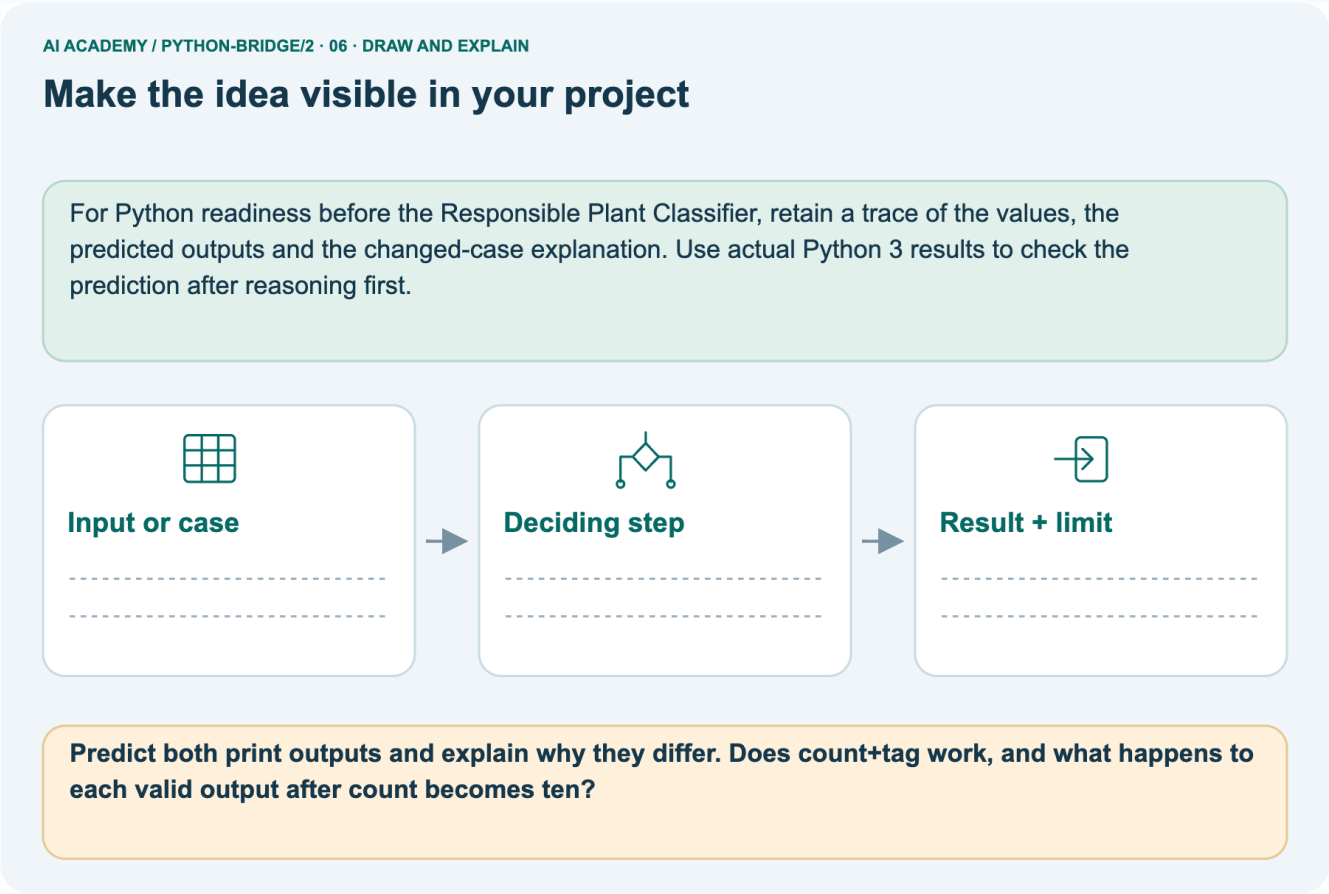
Follow the labelled picture with the matching explanation. Open the vector diagram

Values carry **type**s, not merely appearances. The plus operator performs numeric addition for these integers and concatenation for these **string**s; the mixed operation is invalid here.

Limit: These examples use built-in integers and **string**s; operator behaviour depends on **type**s and is not proof of general model reasoning.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the **value**s, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Predict both print **output**s and explain why they differ. Does count+tag work, and what happens to each valid **output** after count becomes ten?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.